

Fig. 1: The range difference and RF phase components in the GPS carrier frequency at the two receivers from a GPS satellite

known orbits about 20,000 km above the earth and locations on the earth. Once distances are measured, coordinates of positions on the earth are calculated by triangulation. Distances are measured based on the amount of time required for an electromagnetic GPS signal to travel from the satellites to ground-based antennae and receivers.

Satellites transmit precise time and location information in three binary codes: precise code (P code), coarse and acquisi-

tion code (C/A code), and navigation code. The codes are transmitted on two carrier waves that are part of the L-band of the microwave electromagnetic spectrum. The two carrier waves have frequencies called L1 (1575.42MHz, 19cm wavelength) and L2 (1227.60MHz, 24cm wavelength).

Carrier phase is the cumulative number of phase cycles (including the fractional part of the last cycle) of the L1 and/or L2 carrier-wave signal as measured at the receiver. Unlike pseudo-range, use of the carrier phase allows for correction of atmospheric effects and clock offsets between the satellite and receiver. Carrier phase is used for high-precision GPS surveying.

There are mainly two types of GPS surveying methods: kinematic and static surveying. In kinematic applications, receivers are in motion during the measurement period and real-time positioning solutions are available based on the pseudo-range observables. In static applications, receivers are stationary for long measurement periods (generally longer than 30 minutes) and both pseudo-range and carrier-phase data are post-processed for precise positioning solutions. Rapid-static applications are the same as static techniques except that occupation times are short, generally ranging from 5 to 20 minutes, and post-processing relies on code and L1 and L2 carrier-phase observations.

The type and accuracy of positioning in kinematic and static surveying are dependent on the number of receivers available. There are two types of positioning: single-point and relative. Single-point positioning is the determination of a ground position using one receiver and observables from at least four satellites. It relies

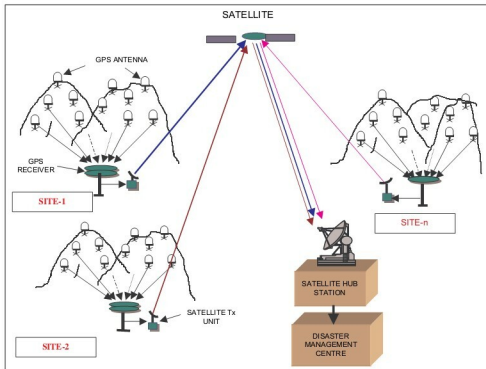


Fig. 2: Satellite-based landslide-monitoring system

monitor all the landslide-prone sites of our country from a central monitoring station and generate an early warning to respective places.

GPS surveying

The primary task of GPS surveying is to measure distances between 24 Department of Defence satellites in

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Two GPS observables are used